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Integrity issues
Reputation issues

Overlay networks

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Outline

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Centralized and distributed systems

- Centralized systems:
  - They are also called single-processor systems.
  - Beginning of computation.
  - Big and expensive computers (mainframes).
  - Few networks to connect them.
  - Independent operation.
- Distributed systems:
  - They appeared due to: (1) Personal Computers (PCs) and (2) Local Area Networks (LANs).
  - Collection of independent computers that appears to its user as a single coherent system.
  - Objective: share resources in a transparent manner for the user (reduce costs, improve scalability, etc.).

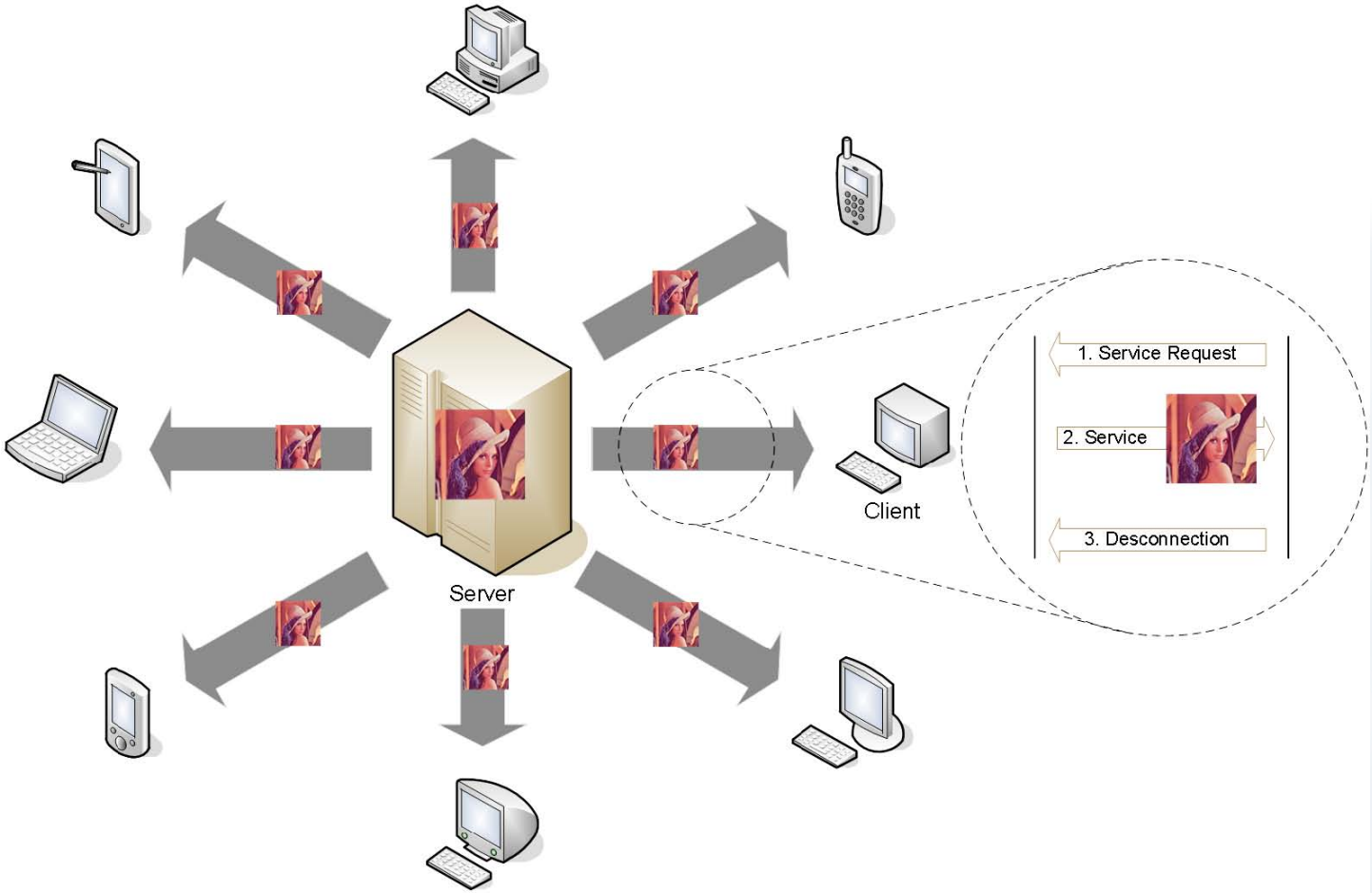
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Types of distributed systems I

- Client/server model:
  - Clients: processes that requests a service from a server.
  - Servers: processes implementing a specific service.
  - Usually, based on request-reply behavior.
  - Few servers provide service to lots of clients.
  - Drawbacks:
    - single point of failure.
    - Server can get overloaded (bottleneck).
    - Generally more expensive and difficult to deploy initially.

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Types of distributed systems II



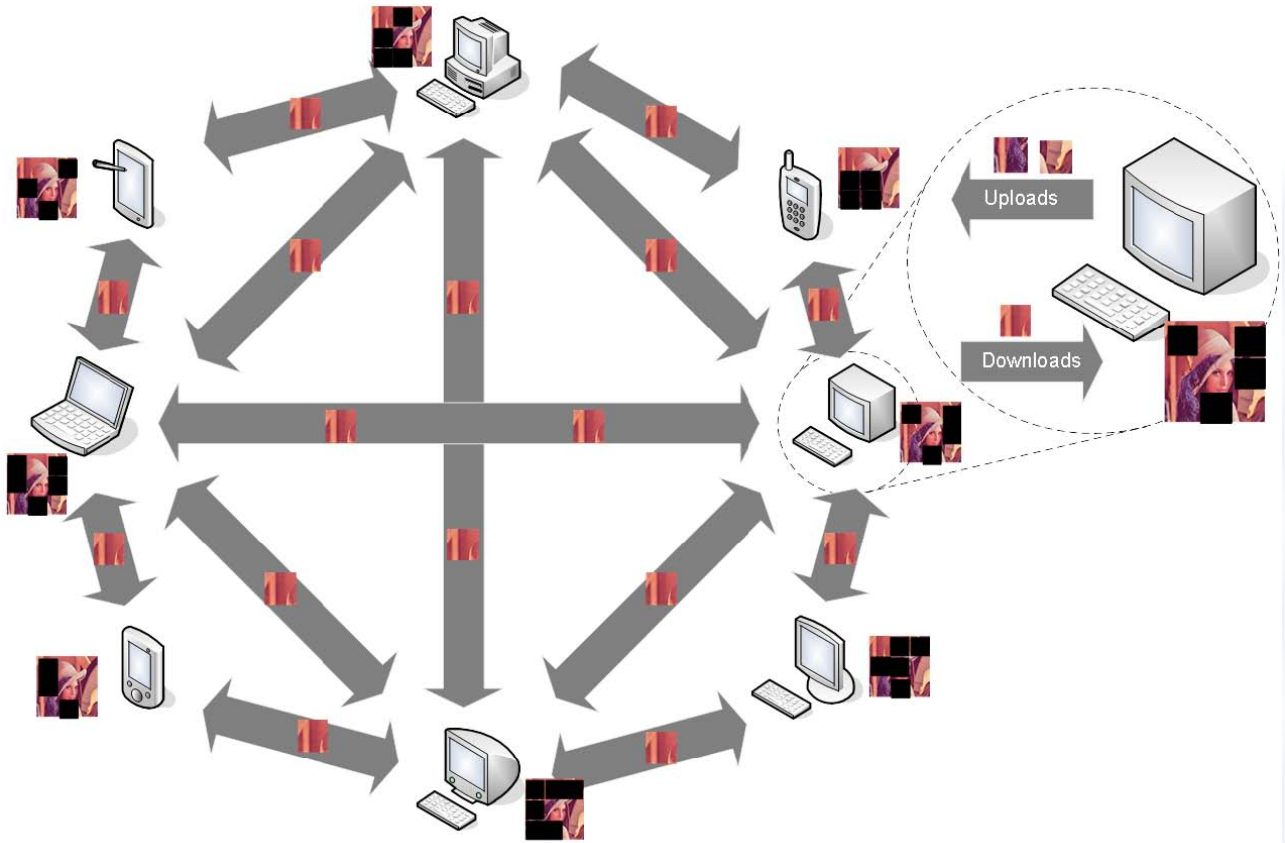
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Types of distributed systems III

- P2P model:
  - Equal and autonomous entities (peers).
  - Nodes contact among them directly.
  - Collaborative applications.
  - Self-organization.
  - Decentralized resource usage.
  - The more peers, the faster the network is.

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Types of distributed systems IV



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P2P networks

- In general, these may be considered p2p networks (in some ways):
  - Mesh networks.
  - MANET.
  - VANET.
  - WSN.
  - Cognitive networks.
  - DTN, etc.
- In particular, the term p2p refers to networks that:
  - Do not care on the specific characteristics of the physical layer, which is always available.
  - Nodes can appear/disappear dynamically.
  - Any number of nodes (from few to millions).

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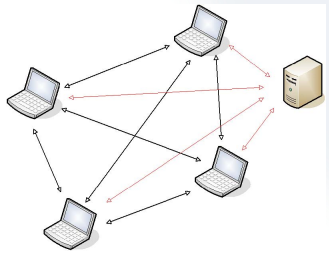
P2P architectures

- Classical:
  - Centralized.
  - Pure (totally decentralized).
  - Hybrid.
- New architectures:
  - Structured P2P overlays.

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Centralized architecture

- Nodes act in a p2p manner, but there is a special node that manages certain functions.
- This server usually maps resources with the peers that have them.
- Advantages:
  - Simplicity.
  - Easiness to locate resources.
- Drawbacks
  - The server is vulnerable (single point of failure).
  - The server could be a bottleneck (scalability).
- Killer app: Napster.



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Napster

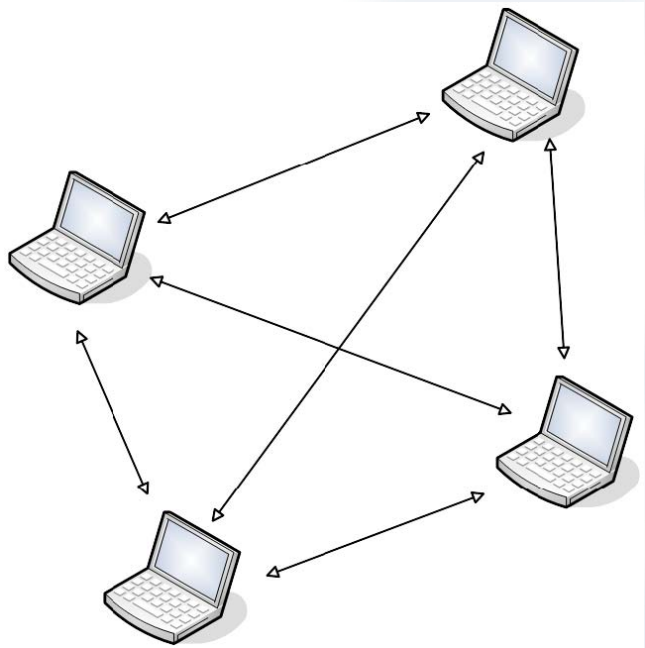
- Music sharing service (mp3) released in 1999, and closed in 2001-2002 due to legal issues.
- It used a centralized server to locate files.
- Download the complete file (not chunks) via ftp.
- The server brokers requests from one peer to another.
- The files did not have to travel through the Napster server because nodes contacted themselves directly.
- DRM (Digital Rights Management).
  - In the analog world, it costs money to make a copy of something.
  - In the digital world, it costs money to prevent copies from being made.
- Learnt lesson: centralize nothing so lawyers can't get you.



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Pure (totally decentralized) architecture

- All nodes are totally equal and have the same functions.
- There is not a single point of failure.
- Flood of messages.
- Examples: Gnutella, Freenet.



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Gnutella

- Distributed searching system.
- It is a communications protocol.
- The Gnutella protocol does not mind how the user will get the resources.
- Download the whole file (not chunks).
- No central element (no legal issues).
  - No single point of failure.
  - Hard to destroy.
- Searches based on flood of messages.
- Unnecessary consumption of bandwidth.



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Freenet

- Bandwidth and disk space sharing.
- Fully decentralized.
- Download the whole file (not chunks).
- Prevent censorship of documents.
- Chain of nodes & hops-to-live count.
- Creates a very large, geographically distributed hard drive.
- Partially structured: files with similar keys tend to create clusters on the same nodes.
- It does not guarantee the data to be found.



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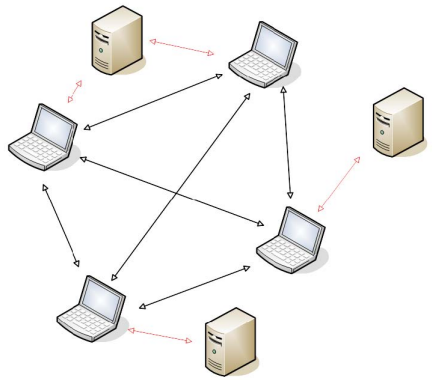
Pure (totally decentralized) architecture

- Drawbacks:
  - There is no guarantee that a certain existing resource will be found.
  - Not scalable due to bandwidth usage.
- Learnt lesson:
  - Some processes perform better in a centralized way.
  - Let us put more central servers (or supernodes).

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Hybrid architecture

- Nodes work in a p2p manner, but there are some nodes that manage certain functions: SUPERPEERS.
- Searching performs better in a centralized way.
- There is no single point of failure.
- Examples: eDonkey, BitTorrent, KaZaA (FastTrack), Skype, etc.



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Edonkey

- File sharing hybrid network.
- Also known as the eDonkey2000 network or eD2k.
- Long term availability of files.
- Based on chunks.
- eDonkey client programs connect to the network to share files.
- eDonkey servers act as communication hubs for the clients, allowing users to locate files within the network.



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BitTorrent

- File sharing service.
- Focused on efficient fetching of files.
- Based on chunks.
- Swarming: download from others downloading same object at same time.
- Searching by means of web pages.
- Distribute the same file to all peers.
- Single publisher (tracker), multiple downloaders.
- Tit-for-tat strategy.



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KaZaA

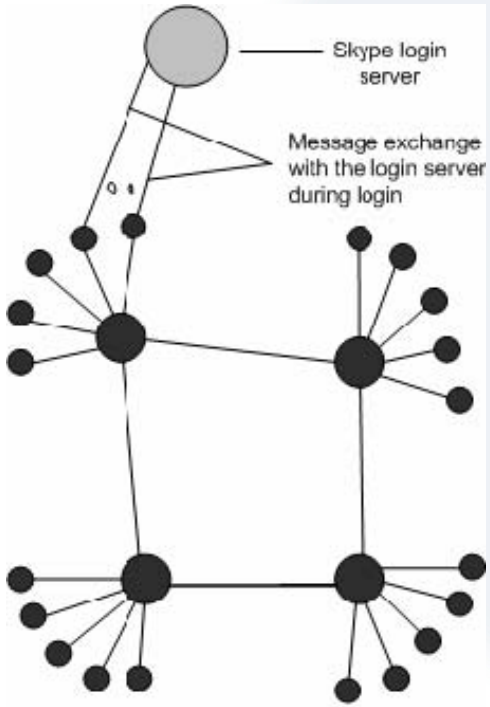
- Peer-to-peer file sharing application.
- Uses the FastTrack protocol.
- Based on chunks.
- Supernodes to improve scalability.
- Swarming: parallel downloading.
- Corruption handling using UUHash: fast but not secure.
- Easy to spread corrupt or fake files.



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Skype

- Software company that offers Voice over IP.
- Proprietary protocol.
- Hybrid network structure.
- Login at Skype server (centralized):
- Further message exchange via P2P network
- P2P network based on Kazaa.
- Provides communications security.



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Hybrid architecture

- Drawbacks:
  - It is possible to attack/close the main servers.
  - In 2006, several edonkey servers (including Razorback2) were confiscated by the Belgian police.
- Learnt lesson:
  - Total availability is only achieved by means of pure (totally decentralized) networks.
  - Let us put order on them!

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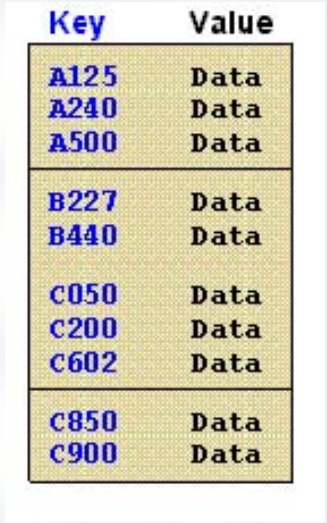
Structured p2p overlays

- Put some order (structure) in a pure p2p network.
- Better design to improve efficiency.
- Totally decentralized.
- Fault tolerant.
- Scalable.
- Self-managed.
- Map resources with their location within the network.
- Allow to find an existing resource in a limited number of hops (routing tables).
- Substrate for high availability and scalability applications.
- Usually, routing decisions are not based on IP addresses but on a Distributed Hash Table (DHT).

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Distributed Hash Tables I

- Hash table distributed among all the nodes of the p2p network.
- They were designed to offer efficient search services.
- They map keys onto values:
  - Values are usually associated to the resources. A value could be a document or an address to get it.
  - Keys are usually associated to the identifier of these resources.
- C. G. Plaxton, R. Rajaraman, and A. W. Richa. Accessing nearby copies of replicated objects in a distributed environment. In Proceedings of the 9th Annual ACM Symposium on Parallel Algorithms and Architectures. 1997.



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Distributed Hash Tables II

- Define an abstract keyspace.
- A partitioning scheme splits ownership of this keyspace among the participating nodes.
- Nodes can find the owner of any given key in the keyspace.
- Example:
  - We want to store a file with name *filename*.
  - The creator obtains the corresponding *key=hash(filename)*.
  - The creator sends a message *put(key,file)*, which is sent through the overlay to the owner of this piece of keyspace.
  - The owner stores the pair *key,file*.
  - Any node that wants to retrieve the file associated to key will send a message *getkey*, which will be routed by the overlay towards the owner.

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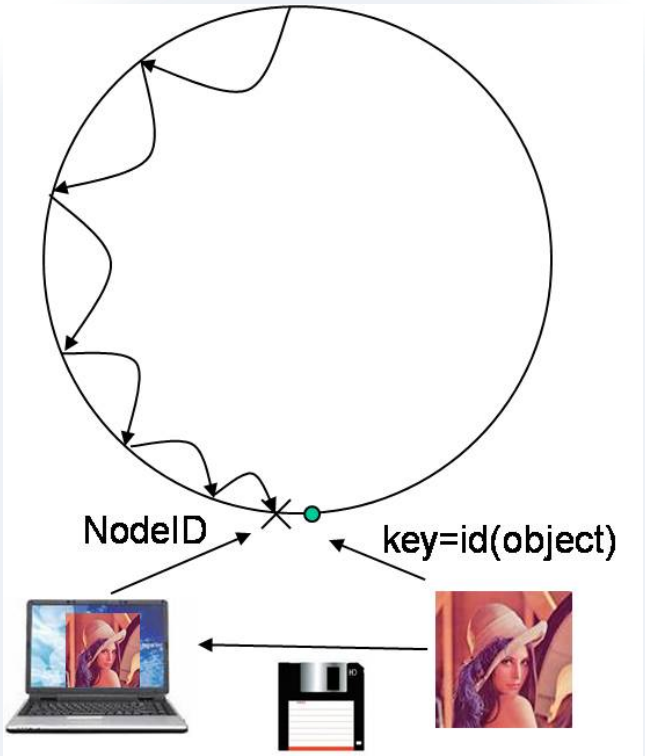
Distributed Hash Tables III

- Issues:
  - Consistent hashing: the addition or removal of one slot within the hash table does not significantly change the mapping of keys.
  - DHTs only support exact-match search, rather than keyword search. Another layer must perform this task.

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Routing in structured p2p overlays I

- Nodes and resources share the same space of directions.
- Each node has an identifier nodeID.
- Each resource has an identifier key.
- A resource is stored in the nodes whose nodeIDs are near the key of the resource.
- We can reach the resource routing messages towards the key.



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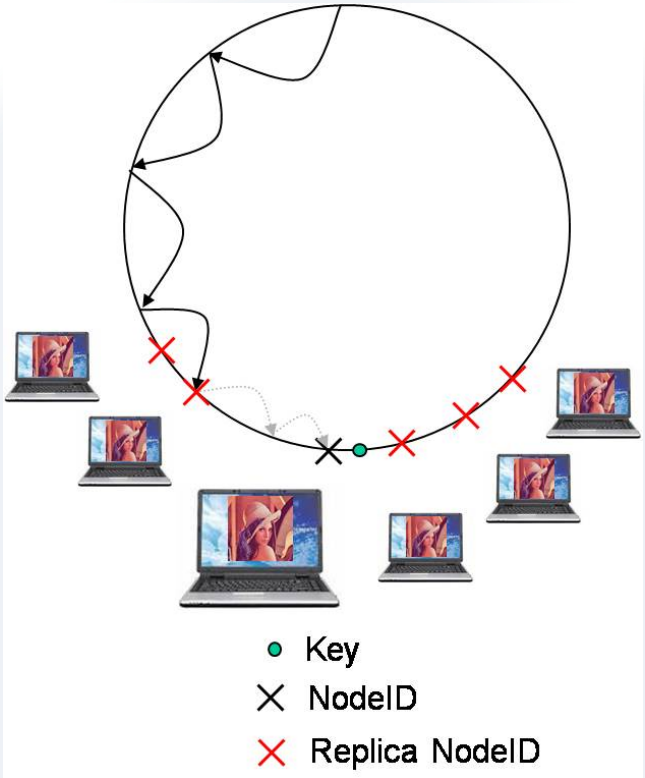
Routing in structured p2p overlays II

- Each node maintains a set of links to other nodes (its neighbors).
- These links form the overlay network, and make routing possible.
- Neighbors are situated according to a certain structure (network topology).

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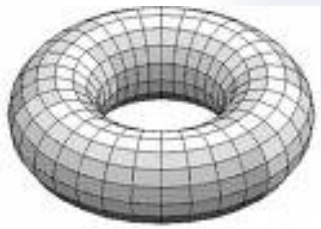
Routing in structured p2p overlays III

- Resources can be stored in more than one place to provide fault tolerance.
- A replication function is defined.
- Replicas are placed near the original object.
- The nearer we are to the key, the more probable to find the original resource or a replica.



CAN (Content Addressable Network)

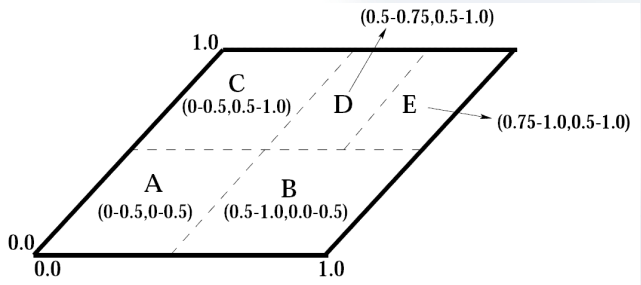
- Scalable indexing mechanism.
- d-dimensional Cartesian coordinate space on a d-torus (totally logical).
- Insertion, lookup and deletion of (key,value) pairs.
- Each node stores a zone (chunk) of the hash table.
- Requests for a particular key are routed by intermediary nodes towards the node whose zone contains the key.
- The network contains the resource.
  Sylvia Ratnasamy, Paul Francis, Scott Shenker, Richard Karp, Mark Handley: “A Scalable Content-Addressable Network”. In Proceedings of ACM SIGCOMM. 2001 (UC Berkeley).



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Routing in CAN

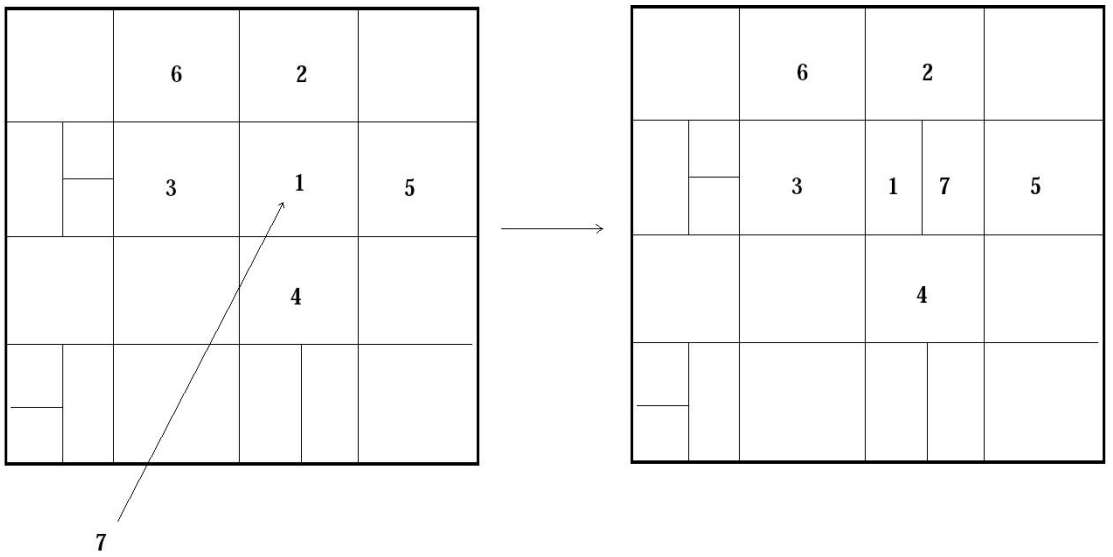
- The entire coordinate space is dynamically partitioned among all the nodes of the system.
- Each node “owns” a zone of the space.
- The space is used to store (key,value) pairs.
- Keys should be known by the user.
- Keys are deterministically mapped onto a point of the space (using a hash function).
- The corresponding (key,value) pair is stored in the node that owns this zone.
- Each CAN node maintains a routing table that holds the IP addresses of the neighbors that manage the closest coordinate zones.



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Bootstrapping in CAN

- New nodes must contact to a bootstrap node.
- The newcomer chooses a random point in the space.
- The bootstrap node uses the routing mechanism to find the node whose zone will be split.
- The newcomer transfers the (key,value) pairs.
- Neighbors are informed about the splitting.



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Node departure in CAN

- Identify a node is departing:
  - Passive departure: via heartbeat messages.
  - Active departure: the departing node helps in the process.
- Zone merging:
  - The released zone is analyzed to determine if a neighbor node can merge it with its own. This will depend on the shape of the zones.
- Update the routing tables.
- Fixing the overlay after a node departure is not an easy task. We will not study this.

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Chord

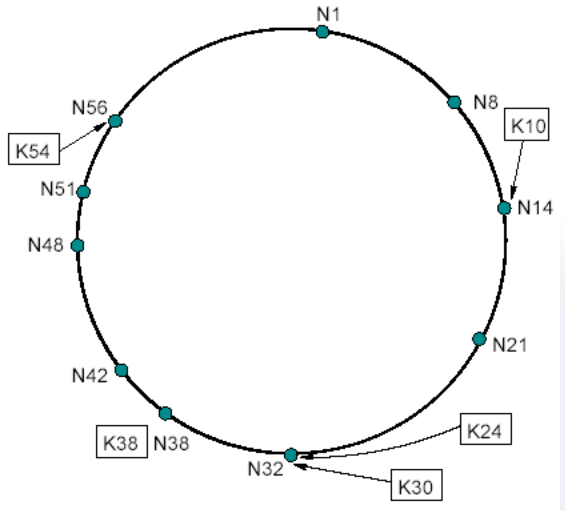
- Distributed lookup service.
- Resources (keys) and nodeIDs are ordered in a 1D circular space modulo 2^m.
- The network contains the resource (not the original node).
- Data location:
  - Associating a key with each value.
  - Storing the (key,value) pair at the successor node of the key.
  - Keys and nodeIDs are uniformly distributed.
  - Hash functions are used to distribute them.

I. Stoica, R. Morris, D. Karger, M. F. Kaashoek, and H. Balakrishnan: “Chord: A Scalable Peer-to-peer Lookup Service for Internet Applications”. In Proceedings of SIGCOMM. 2001 (MIT).

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Routing in Chord I

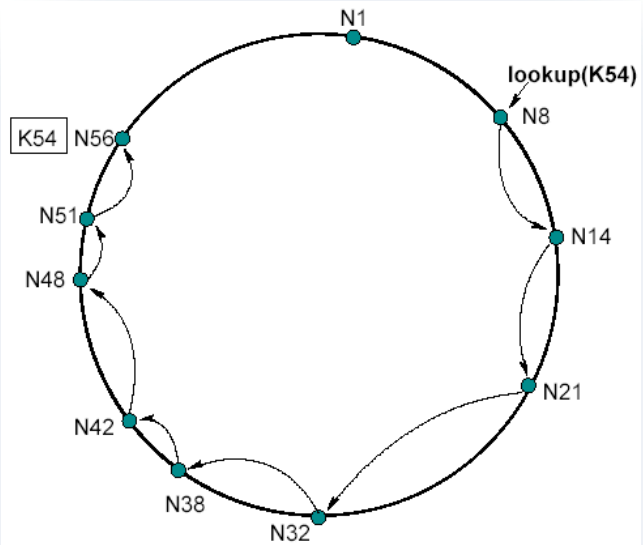
- Identifier circle of m bits.
- m should be large enough to make collisions improbable.
- NodeIDs and keys are labeled 0, 1, 2, . . . , 2^m-1.
- Each node has a successor (next node clockwise) and a predecessor (next node counter-clockwise).
  - Example: m=6
    - NodeIDs={1,8,14,21,32,38,42,48,51,56}
    - Keys={10,24,30,38,54}
  - The pairs key,value are stored in node successor (key), which is the node whose nodeID is equal to or follows the key.



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Routing in Chord II

- Each node knows the IP address of its successor, so it is possible to perform a linear search over the network for a particular key (naïve method).
- Extra routing information can be used (finger tables) to avoid traversing all nodes.
- Nodes know more about other nodes closely.



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Pastry

- Object location and routing.
- Each node has a unique identifier (nodeID) within a circular keyspace.
- Pastry routes the message to the node with the nodeID closest to the key.
- Similar to Chord.
- Differences:
  - Resources are located in the original node.
  - Takes into account network locality.
    - External routing metrics (ping or traceroute) to determine the best routes to store in its routing table.
    - NodeIDs are sub-divided into levels to specify domains.
    - Shortest hop count, lowest latency, etc.

A. Rowstron and P. Druschel. "Pastry: Scalable, decentralized object location and routing for large-scale peer-to-peer systems". IFIP/ACM International Conference on Distributed Systems Platforms (Middleware). 2001 (Microsoft Research).

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Routing in Pastry

- Routing based on:
  - A list of leaf nodes: a list of the closest peers by nodeID in each direction around the circle.
  - A neighborhood list: closest peers in terms of the routing metric.
  - A routing table: routing based on prefixes. The routing table contains the address of the closest known peer for each possible digit at each address level.

- Nodes forward the message to a node whose nodeID shares with the key a prefix that is at least one digit longer.



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Tapestry I

- Resources are located in the original node, and Tapestry publishes them.
- To alleviate the problem of single point of failure, all resources are replicated in other nodes (surrogate nodes).
- But the owner is the unique root node for every (key,value) pair.
- Similar to the original DHT of Plaxton.
- Messages keep on going to the original root node (routing based on prefixes).

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Tapestry II

- The route can be interrupted in case of finding a copy of the desired resource in a surrogate node.
- A node has a neighbor map with multiple levels and neighbor hops are chosen for network locality.

Zhao, Kubiatowicz, Joseph. Tapestry: “An Infrastructure for Fault-tolerant Wide-area Location and Routing”. Technical Report Identifier: CSD-01-1141. 2001.

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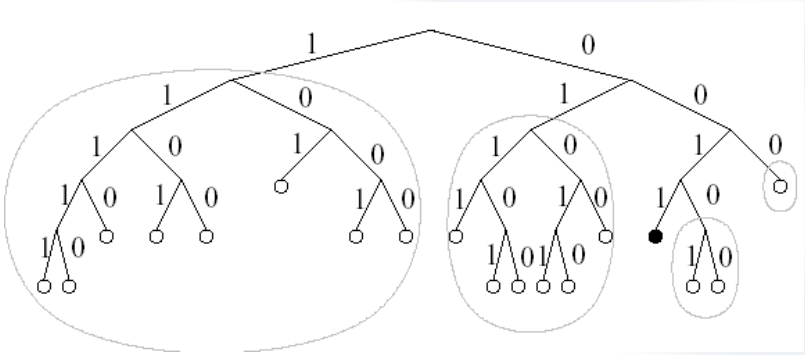
Kademlia

- Binary tree structure.
- Keys and nodeIDs have the same length (160 bits) and format (generated by a hash function).
- Communications among nodes using UDP.
- The network can store the contents, or can provide the location of them.
- The Kad network uses (a variant of) Kademlia.
- In Kad, usually, each node re-publishes (key,value) pairs as necessary to keep them alive.

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Routing in Kademlia I

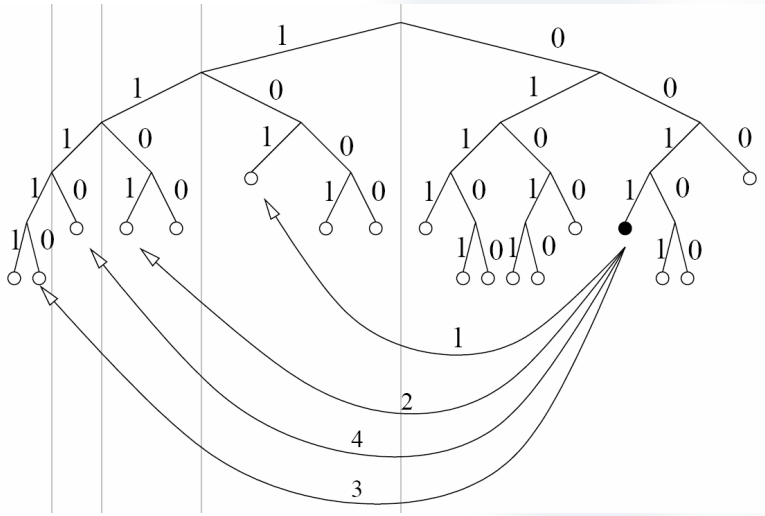
- List for each bit of its nodeID (k-buckets).
- There are typically up to k=20 contacts in each list.
- Entries are node locators in the form (IP, port, nodeID).
- Routing is based on the distance between nodes (exclusive or).
- Every list corresponds to a specific distance from the node.
- The lists are filled dynamically during transactions.
- Binary tree structure organized according to the nodeIDs.



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Routing in Kademlia II

- To find a particular node (or key):
- Query to the k nodes contained in the corresponding list (based on the start of the node/key). These are the known nodes closest to it.
- These nodes return the k closest nodes that they know.
- Select best k, and iterate.



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JXTA

- JXTA is a set of open protocols that enable the creation and deployment of p2p networks.
- Standardize the manner in which peers:
  - Discover each other
  - Self-organize into peer groups
  - Advertise and discover network services
  - Communicate with each other
  - Monitor each other
- JXTA is a middleware, designed as a set of building blocks to allow developers to rapidly develop P2P applications.

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Security issues I

- Architectures often suppose that nodes will act honestly.
- This is not acceptable in open systems like the Internet.
- Anonymity adds extra problems.
- Pure architectures do not have centralized authorities that monitor or punish malicious peers.
- Some of the most important problems are:
  - How we identify nodes?
  - How we route messages securely?
  - How we assure that contents have not been corrupted?
  - How we choose the best server provider?
  - How we assure privacy of my personal data?

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Identity and access control I

- Centralized management:
  - Registration in a server.
  - This server is responsible for issuing the identifiers of the nodes.
  - This server may ideally perform the access control.
  - Generally, servers usually accept any newcomer.
  - Problems to provide anonymity.
  - May be impossible in pure p2p systems.

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Identity and access control II

- Distributed management:
  - No server at all.
  - Self-managed identities.
  - Anonymity can be provided.
  - Newcomers must know some internal node to access the network.

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Types of identities

- Based on IP addresses (for instance, hashing it):
  - Verifiable.
  - Stability: not easy to change (unless IPv6).
  - Problems when IP addresses are dynamic.
  - IP-spoofing.
  - Poor level of anonymity.
- Based on public keys (for instance, hashing it):
  - With a certificate (not self-managed):
    - Verifiable.
    - Stable.
    - May not be anonymous.
  - Without a certificate (or self-signed certificated):
    - Not verifiable
    - Not stable.
    - May provide anonymity.
- Based on other random numbers:
  - Not verifiable.
  - Anonymous.

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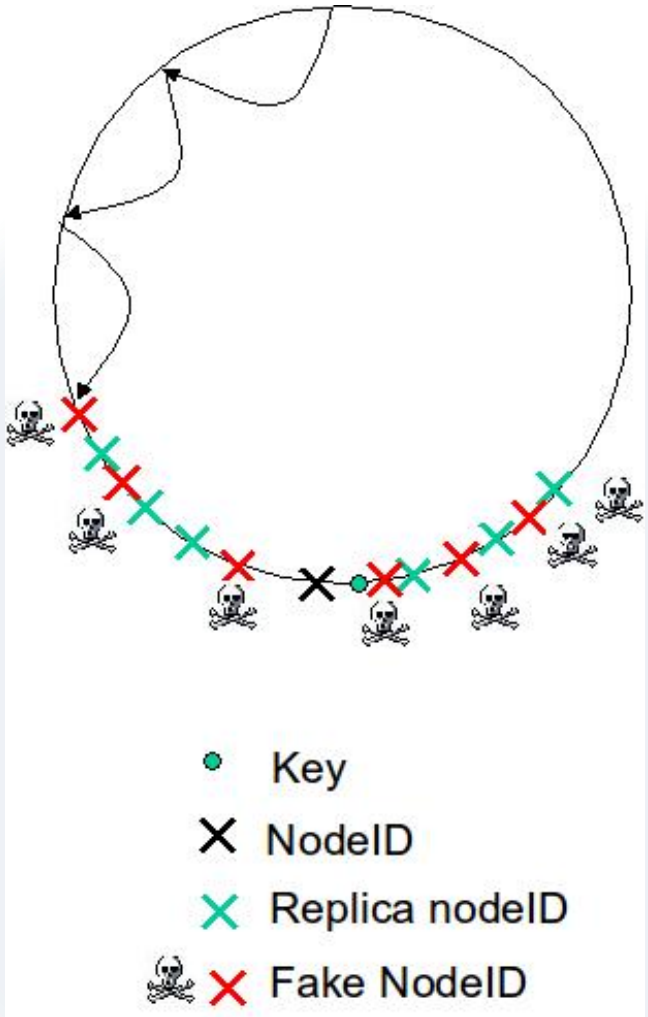
Identity and access control

- Gnutella:
  - You access the network by connecting to one arbitrary node.
  - You choose a random identifier.
- Edonkey/Kad:
  - You access the network by connecting to one server.
  - When the client is used for the first time automatically chooses a random number (userhash).
  - Stored in preferences.dat (or userhash.dat).

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Identity in P2P overlays

- Specially dangerous in structured overlays.
- In most overlays the location of the nodes is directly related to their nodeID.
- A malicious node could choose its identity in a malicious way:
  - It can try to supplant another node.
  - It can obtain multiple identities (Sybil attack).



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Identity and access control in P2P overlays I

- CAN:
  - There are no identities, but nodes are identified by the zone they manage.
  - Newcomers choose their zones every time they join the network.
  - This way of proceeding causes multiple security problems.
  - Drawbacks:
    - NodeIDs are self-managed. Malicious nodes can censor (manage) a target document.
    - NodeIDs are not stable. Malicious nodes can disrupt the network without paying any cost.
    - CAN is vulnerable to Sybil attacks. Malicious nodes controlling several zones can corrupt the routing system.

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Identity and access control in P2P overlays II

- Chord:
  - Node's identifier (nodeID) is chosen by hashing the node's IP address.
  - The network only checks that this nodeID is unique.
  - The newcomers can join and leave the network at any time.
  - The authors do not define any mechanism for access control.
  - Drawbacks:
    - NodeIDs are not stable. Most users use dynamic addresses.
    - NodeIDs are partially self-managed. IP addresses are easily supplanted (IP-spoofing).
    - Using IP addresses to generate nodeIDs can cause the loss of anonymity.
    - Chord is also vulnerable to Sybil attacks.

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Identity and access control in P2P overlays III

- Pastry:
  - Each node has a unique identifier (nodeID) of 128 bits.
  - NodeIDs can be generated by a high-quality uniform random number generator or a secure hash function.
  - The authors did not describe how access control is performed.
  - Any newcomer is just accepted as a valid peer becoming part of the Pastry network.
  - Drawbacks:
    - NodeIDs can be self-managed and not stable. The users can change their IP addresses deliberately.
    - Using IP addresses to generate nodeIDs can cause the loss of anonymity.
    - Pastry is also vulnerable to Sybil attacks.
    - Without a access control mechanism it is very difficult and ineffective to use reputation systems.

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Identity and access control in P2P overlays IV

- Tapestry:
  - Tapestry uses SHA-1 to produce a 160-bit identifier.
  - The newcomer must generate its own identifier, but the authors do not define how.
  - They do not define an access control mechanism.
  - Tapestry has the same security weaknesses than Pastry or Chord.

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Identity and access control in P2P overlays V

- Kademlia:
  - Node's idenfier is a random number.
  - In Kademlia, like previous ones, the nodeIDs are self-managed and not stable.
  - The authors did not describe how access control is performed.
  - A user can easily perform the Sybil attack or reputation systems have little utility, among others problems.
- JXTA:
  - End-users join the network by authenticating using a username and password.
  - The authors do not define any other type of security measure.

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Identity and access control in P2P overlays VI

- Concluding remarks:
  - Trusting the identity of users to their IP addresses can cause problems.
    - IP addresses are vulnerable to IP-spoofing, and often are dynamic.
    - Readmission of malicious nodes by simply changing their IP addresses.
    - Poor level of anonymity (even hashing).
  - Some networks also allow nodes to use self-managed identities.
    - Malicious nodes can generate a new identity each time they behave badly.
  - It is necessary to control who acquires an identity. Malicious nodes may acquire a large number of nodeIDs and generate Sybil attacks.

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Routing problems

- Most architectures suppose that nodes will follow truthfully the routing algorithm.
  - Hostile nodes could focus their attacks on:
  - Routing tables management: nodes could exchange fake routing information.
  - Secure message routing: nodes could route incorrectly, corrupt or delete messages.

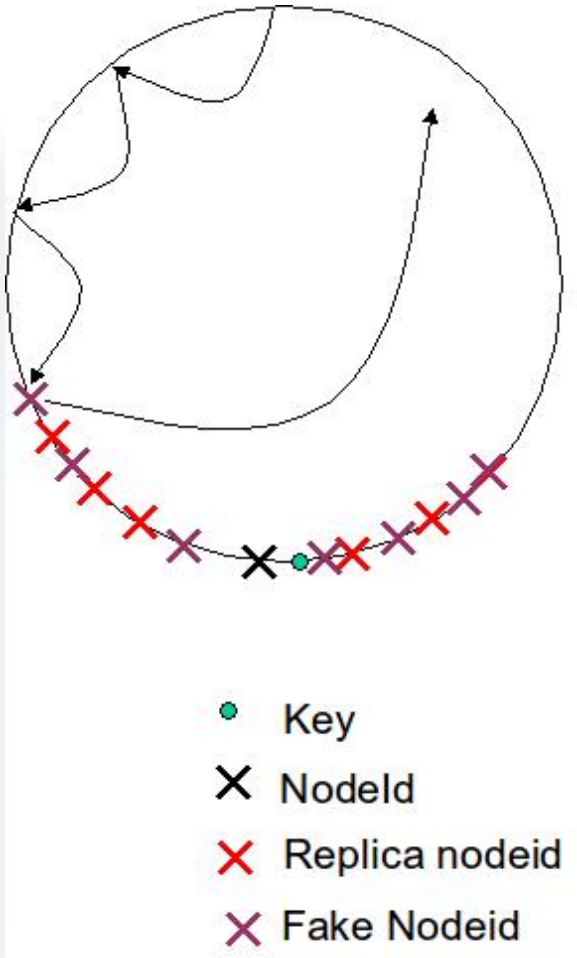
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Routing tables management

- Some systems adapt the routing information to the network conditions.
- Malicious nodes could lie about these network conditions.
- Nodes could send update messages with wrong routing information.
- Solution: system that measure the validity of routing information (Reputation?).
- Routing should be randomized because protocols usually behave in the same manner.

Secure transmission of messages

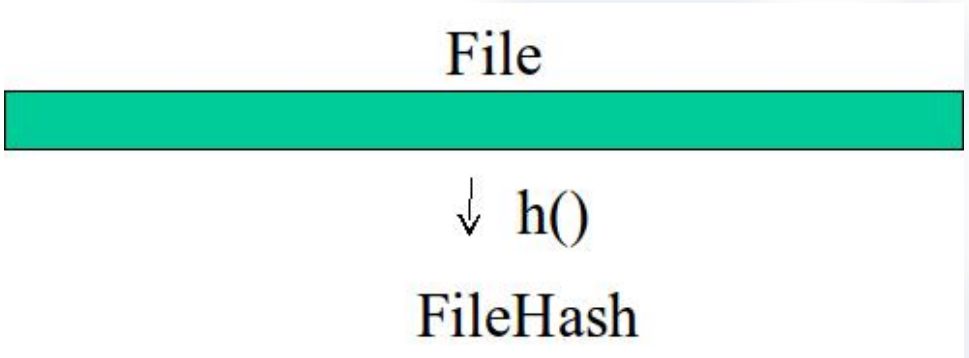
- Routing is a best-efford service.
- An hostile node could break the routing algorithm
- It could corrupt or delete messages.
- Solution: system that avoids or reduce the possibility of breaking the routing of messages (Reputation again?).



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Data integrity

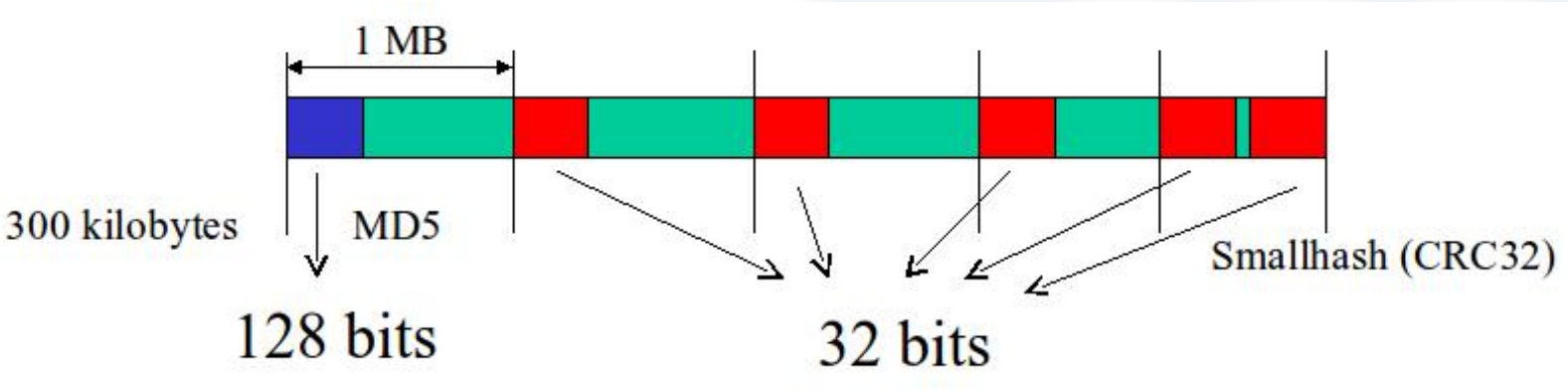
- An adversary could substitute, corrupt or modify data.
- Users must know if data have been modified.
- There is no entity that certifies the integrity of data.
- Anonymity adds extra difficulties.
- Solution: autoverification techniques.
- The FileHash could be used to detect corruptions.
- Problems:
  - Download the whole file to verify correctness. Problems with big files: chunks.
  - How to publish this information. In the key?



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KaZaA/FastTrack: UUHash

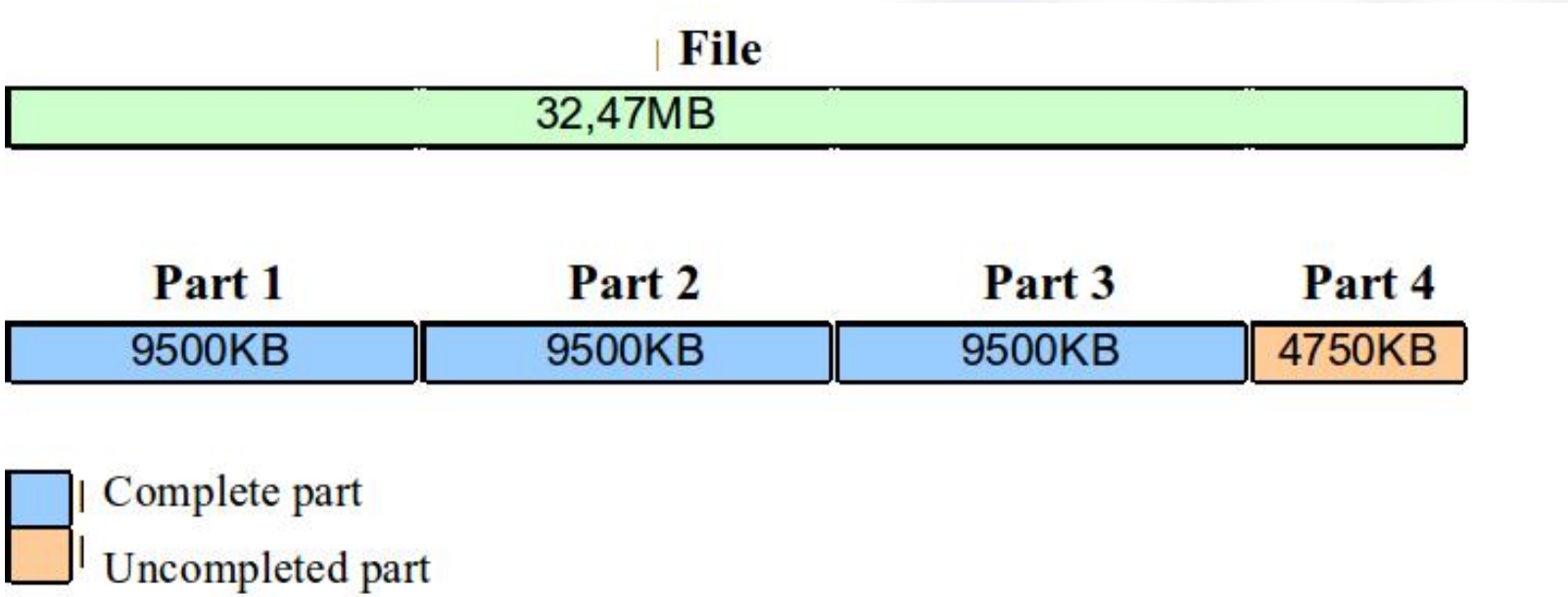
- Allows to verify the integrity of large files.
- Only applies the hash over a fraction of the file.
- There are parts whose integrity is not assured.
- It is trivial to create hash collisions.



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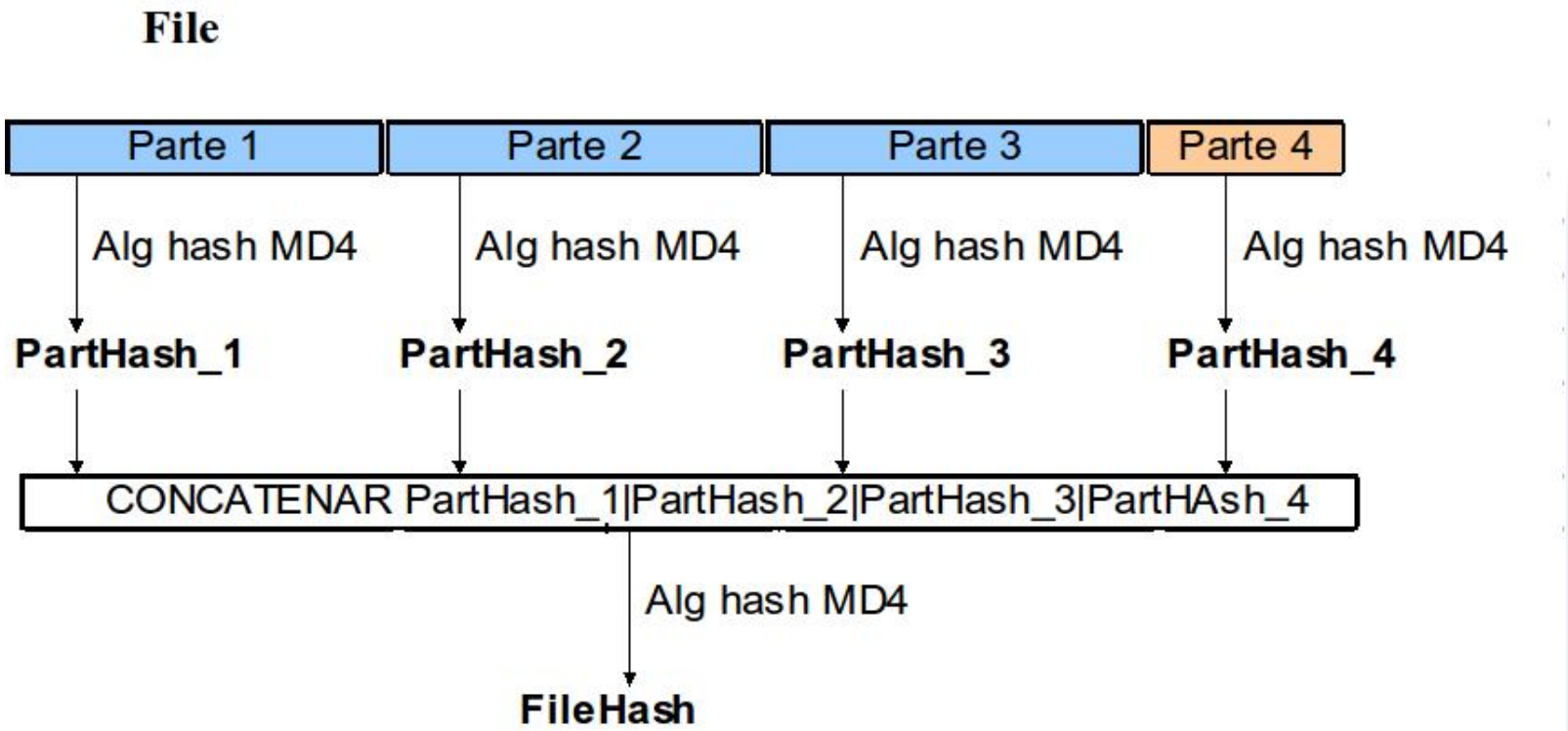
Kad and Edonkey I

- File partitioning:
  - Improve file distribution.
  - Corruption handling.



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Kad and Edonkey II

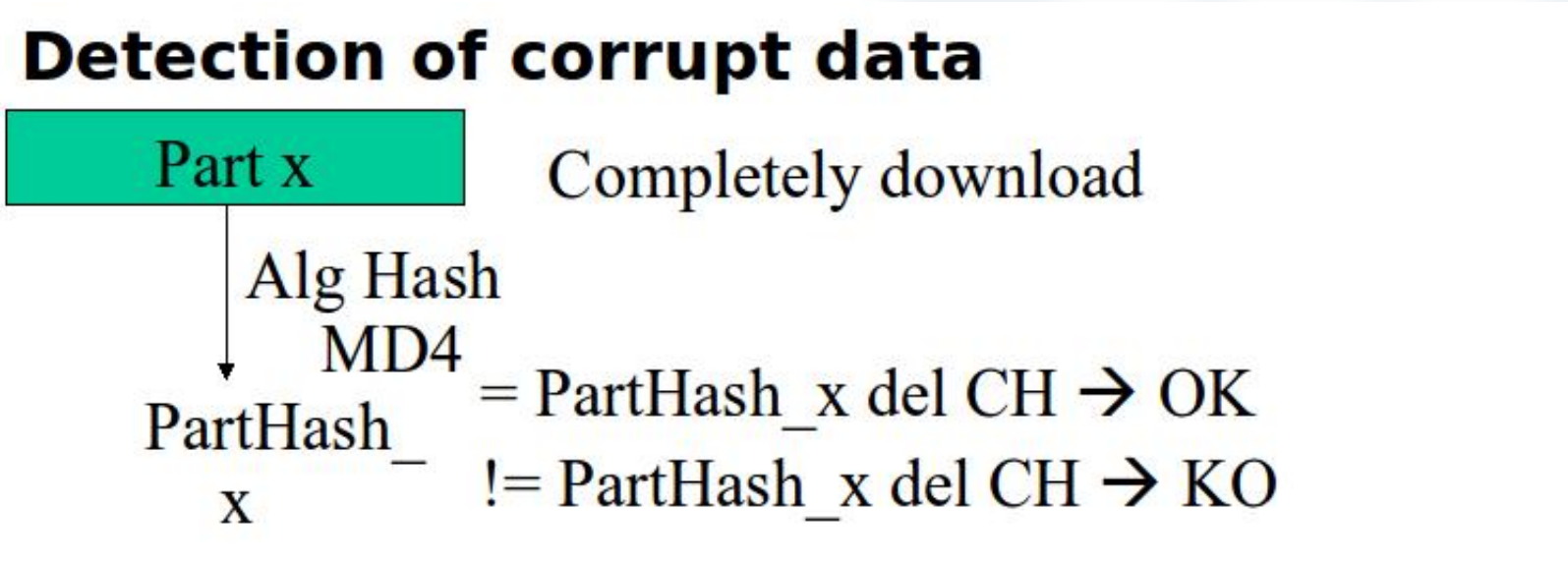


ed2k://|file|myfotos.zip|65164328|BC2A5132A87689FC9A1DBABC7459FBA2|/

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Data integrity in eDonkey I

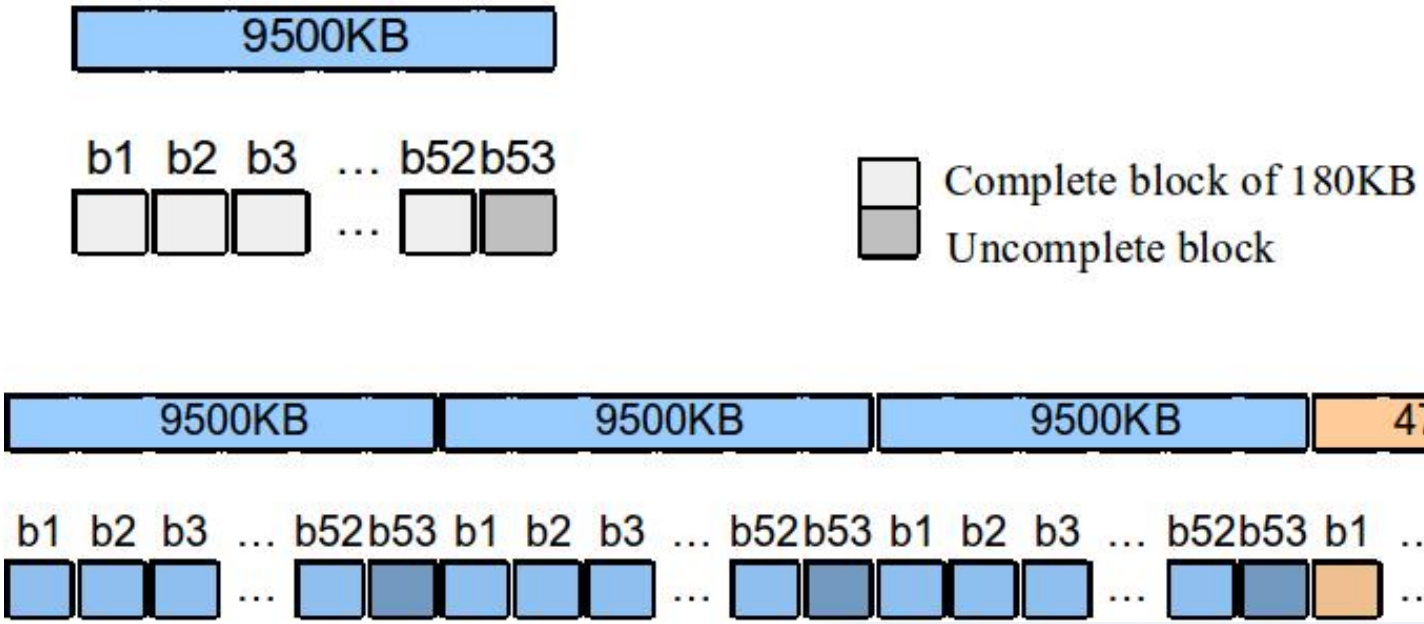
- eMule have two mechanisms:
  - ICH: simple
  - AICH: a little more sophisticated, based on ICH.



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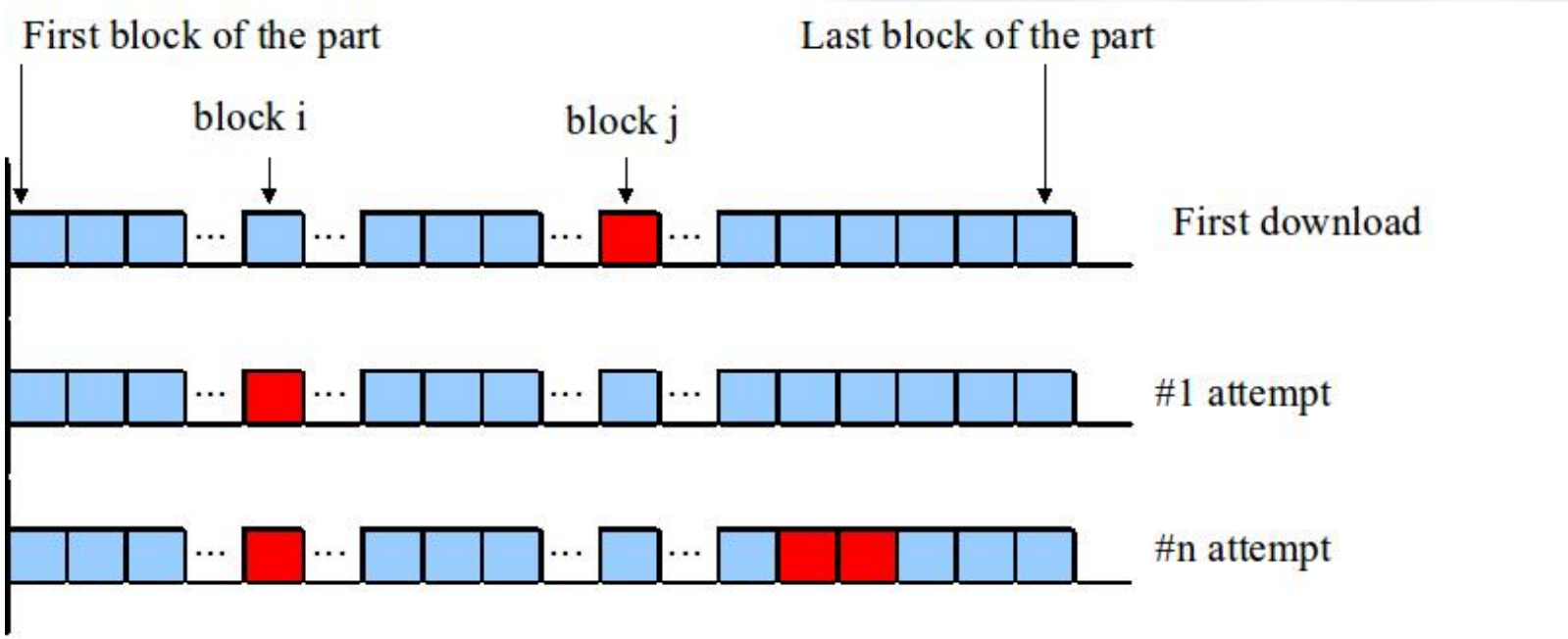
Data integrity in eDonkey II

Parts are divided in blocks:



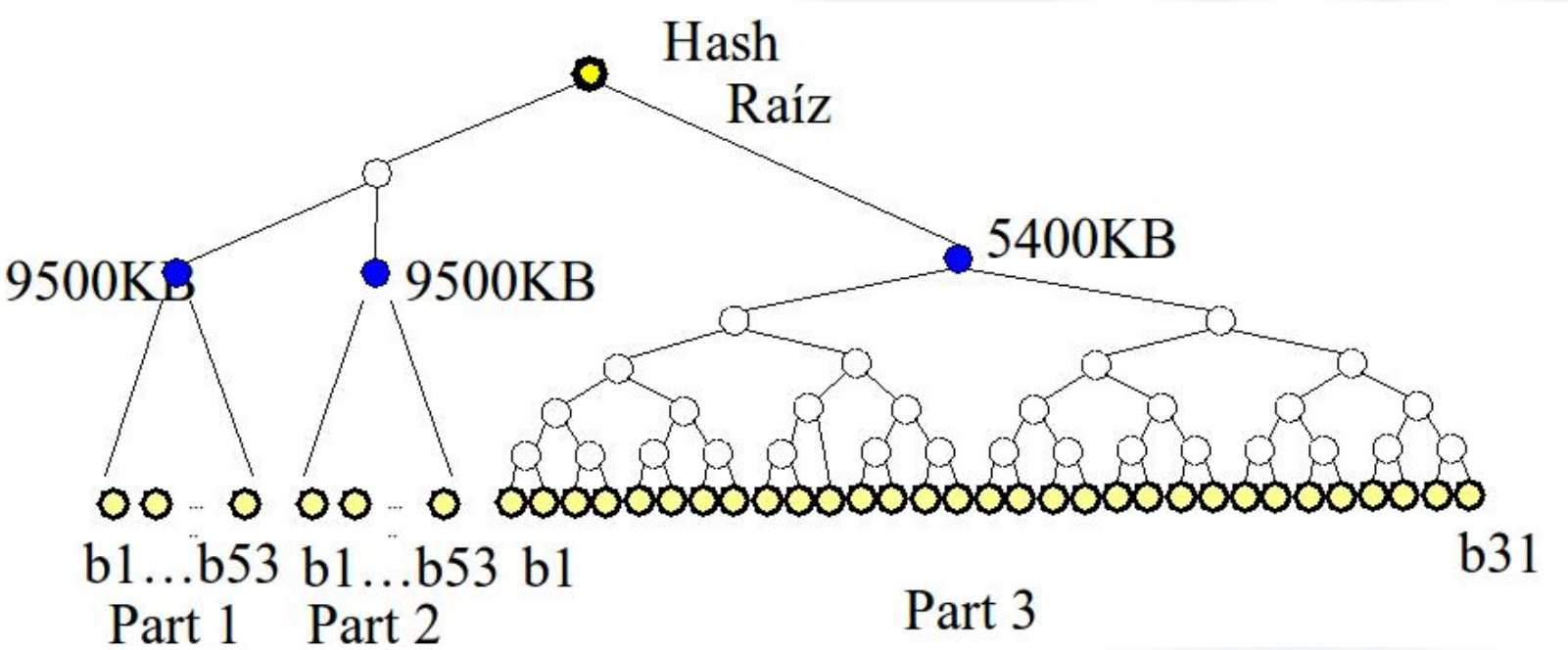
ICH (Intelligent Corruption Handling)

- If a part is corrupt, it starts downloading block by block.
- Each newly downloaded block substitutes the old one, and integrity is verified.
- Inefficient if high corruption probability.



AICH (Advanced Intelligent Corruption Handling)

- Identify correct blocks.
- AICH defines a hash tree, block hashes and root hash.
- Working:
  - Detect corrupt part.
  - Obtain correct block hashes.
  - Identify corrupt blocks.
  - Download corrupt blocks again.



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Data integrity in structured p2p overlays

- The hash of the object could be part of the key.
- Nodes should know (by any external mechanism) the key of the object they are looking for.
- Integrity is assured if the hash of the asked content equals the key.

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Selecting the service provider

- How do we choose the best service provider in p2p systems?
- No authorities to help in this process.
- Anonymity may introduce extra problems.
- Solution: trust and reputation (social control).
- These systems could be used in any decision making, not only to decide the best service provider.
- Problem: trust and reputation depends on the stability of identities.

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Trust

- Expectation that a certain agent will act in a certain way when performing an action.
- We are able to increase trust by reducing the number of people to be trusted.
- We can increase trust by reducing risk.
- There are many definitions of trust (this is one):
- Trust (or, symmetrically, distrust) is a particular level of the subjective probability with which an agent assesses that another agent or group of agents will perform a particular action, both before he can monitor such action (or independently of his capacity ever to be able to monitor it) and in a context in which it affects his own action.

  Gambetta, Diego (2000) 'Can We Trust Trust?', in Gambetta, Diego (ed.) Trust: Making and Breaking Cooperative Relations, electronic edition, Department of Sociology, University of Oxford, chapter 13, pp. 213-237.
- How can be measure trust?

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Reputation

- Memory and summary of behavior from past transactions.
- Trust is often increased by having positive reputation.
- The possible effect on one's reputation also influences how one behaves.
- Reputation depends on the stability of identities.
- One possible definition:
  - Reputation is an expectation about an agent's behavior based on information about it or observations of its past behavior.
    Supporting Trust in Virtual Communities. Alfarez Abdul-Rahman and Stephen Hailes. In Proceedings of HICSS '00, the 33rd Hawaii International Conference on System Sciences-Volume 6.
- We will focus on reputation systems for p2p networks.

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Reputation in p2p I

- P2p systems ideally:
  - All nodes act as equals and unselfishly.
  - All nodes act honestly.
- P2p actually:
  - 10 % of nodes serve 90 % of requests.
  - Nodes obtain resources without any collaboration (freeRiders, leechers).
  - Nodes introduce fake contents in the network.
- Reputation could be a mechanism to avoid malicious behaviors and to promote honest collaboration among peers.

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Reputation in p2p II

- Possible uses of reputation:
  - Distinguish most reputed service providers.
  - Punish improper and selfish behaviors.
  - Decide which peers can connect to the network.
  - Promote stable identities.
  - Promote honest cooperation.
  - Maintain the overall credibility of the system.
  - Etc.

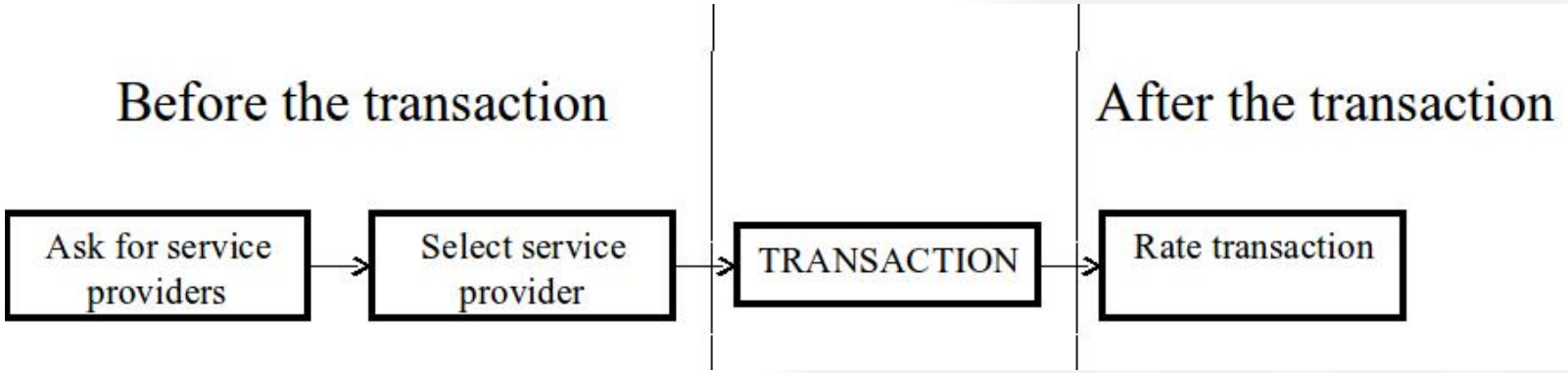
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Reputation in p2p III

- System that obtains, stores, uses and distributes reputation information.
- Desirable properties:
  - Distributed topology.
  - Scalable.
  - Robust against attacks
  - Maintain anonymity (if required).
  - Promote stable identifiers.
  - Multifaceted dynamical reputation (service-aware).
  - Able to distinguish bad from new reputation.
  - Time-descending: old reputation.
  - Reciprocity: good peers take advantage of their behavior.

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Reputation model



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Ask for service providers

- Service Discovery Protocol.
- SDP should be provided by the underlying network.
- The result is a list of service provider candidates.

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Select service provider I

- INFORMATION GATHERING:
  - First hand information (local):
    - Own experiences.
    - Present and past behavior.
    - Trusted source of information.
  - Second hand information:
    - Other's recommendations.
    - From trusted entities (for instance, a CA). Trusted information.
    - From non trusted entities (for instance, an unknown peer). Credibility.

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Select service provider II

- Information we gather or ask?
  - Reputation score directly.
  - Number of sat and unsat transactions.
  - Time information.
  - All history/recent events.
  - Distribution of recommendations:
  - No distribution.
  - Polling (voting).
  - Certificates
  - Etc.

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Select service provider III

- Scoring:
  - Methods:
  - Analytic expressions.
  - Bayesian networks.
  - Fuzzy logic.
  - Game theory.
  - Bio-inspired algorithms.
  - Etc. . .
- Levels:
  - Satisfactory or unsatisfactory.
  - Satisfaction levels.
  - Labels: best reputation, good reputation, normal reputation, bad reputation, etc.
  - Scaled integers:1-10.
  - Probability: [0,1].
  - Aggregation algorithm.
  - Coherency between scores

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Select service provider IV

- RANKING AND SELECTION:
  - Best reputation score?
  - Newcomers?
  - Availability?
  - More resources?
  - Proximity?
  - Etc. . .
  - Mixed criteria?

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Identity issues
Routing issues
Integrity issues
Reputation issues

Transaction

- Service is provided with a certain QoS.
- Is the same QoS than promised in SDP?

P2P
Introduction to p2p networks
Centralized architectures
Pure architectures
Hybrid architectures
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Rate transaction

- Requires human interaction?
- PUNISH AND REWARDING ACTIONS:
  - Local or global actions.
  - Verifiability.
  - Redemption of bad peers.
- HOW TO UPDATE THE REPUTATION DATABASE:
  - Historical or current behavior.
  - Send recommendations to others?